Second-Order System Response

Damping ratio and natural frequency set the whole transient.

Ogata, *Modern Control Engineering*, Ch. 5.

In this tutorial you will:

- sweep the damping ratio ζ and watch the response change,
- compute rise time, peak time, overshoot, and settling time, and
- check the closed-form specs against `stepinfo`.

Step through with **Ctrl+Enter**, or render a report with `publish`.

The standard second-order transfer function is

$\frac{Y(s)}{R(s)} = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$

where ω_n is the **undamped natural frequency** and ζ is the **damping ratio**. The closed-loop poles are

$s = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$

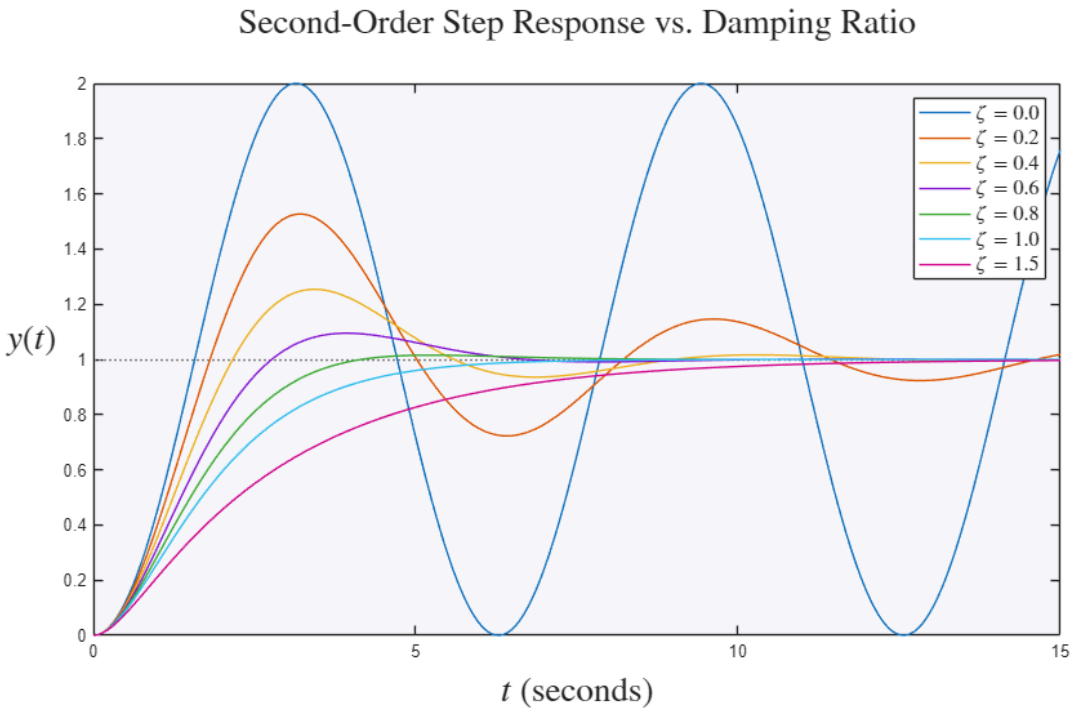
Depending on ζ, the system is:

- **Overdamped** ($\zeta > 1$): two distinct real poles, no overshoot.
- **Critically damped** ($\zeta = 1$): repeated real pole at $-\omega_n$.
- **Underdamped** ($0 < \zeta < 1$): complex-conjugate poles, decaying oscillation.
- **Undamped** ($\zeta = 0$): poles on the $j\omega$-axis, sustained oscillation at ω_n.

Contents

Family of Step Responses for Varying ζ

```
wn = 1;
zetas = [0 0.2 0.4 0.6 0.8 1.0 1.5];
figure
hold on
for z = zetas
    Gz = tf(wn^2,[1 2*z*wn wn^2]);
    step(Gz, 0:0.01:15)
end
hold off
legend(arrayfun(@(z) sprintf('$\\zeta=%.1f$',z), zetas, 'UniformOutput', false), ...
    'Interpreter','latex','FontSize',12)
title('Second-Order Step Response vs. Damping Ratio','Interpreter','latex','FontSize',20)
ylabel('$y(t)$','Interpreter','latex','FontSize',20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$','Interpreter','latex','FontSize',20)
```



Underdamped Transient-Response Specifications

For the underdamped case ($0 < \zeta < 1$), the unit-step response is

$$y(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_d t + \beta), \quad \omega_d = \omega_n\sqrt{1-\zeta^2}, \quad \beta = \cos^{-1}\zeta$$

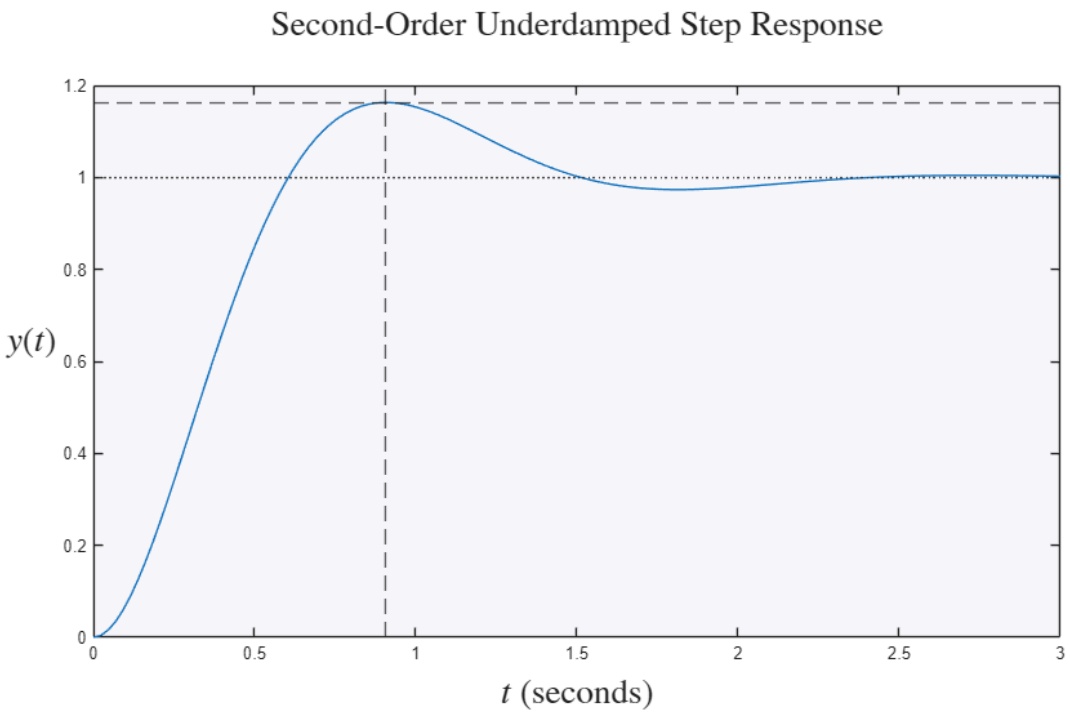
From this, the standard transient specifications are derived:

- **Rise time** (0% to 100%): $t_r \approx \frac{\pi-\beta}{\omega_d}$
- **Peak time**: $t_p = \frac{\pi}{\omega_d}$
- **Maximum overshoot**: $M_p = e^{-\zeta\pi/\sqrt{1-\zeta^2}}$ (fraction of final value)
- **2% Settling time**: $t_s \approx \frac{4}{\zeta\omega_n}$

```
zeta = 0.5; wn = 4;
G = tf(wn^2,[1 2*zeta*wn wn^2]);

wd = wn*sqrt(1-zeta^2);
beta = acos(zeta);
tr = (pi-beta)/wd;
tp = pi/wd;
Mp = exp(-zeta*pi/sqrt(1-zeta^2));
ts = 4/(zeta*wn);

fprintf('Rise time      tr = %.4f s\n', tr)
fprintf('Peak time      tp = %.4f s\n', tp)
fprintf('Max overshoot  Mp = %.2f%%\n', Mp*100)
fprintf('Settling time  ts = %.4f s (2%% criterion)\n', ts)

figure
step(G)
hold on
yline(1+Mp,'--')
yline(1,':')
xline(tp,'--')
hold off
title('Second-Order Underdamped Step Response','Interpreter','latex','FontSize',20)
ylabel('$y(t)$','Interpreter','latex','FontSize',20)
```

```
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$','Interpreter','latex','FontSize',20)
```

```
Rise time      tr = 0.6046 s
Peak time      tp = 0.9069 s
Max overshoot  Mp = 16.30%
Settling time  ts = 2.0000 s (2% criterion)
```



Verification via MATLAB's stepinfo

The Control System Toolbox's `stepinfo` computes these same specifications numerically from the simulated response, providing a check on the closed-form formulas above.

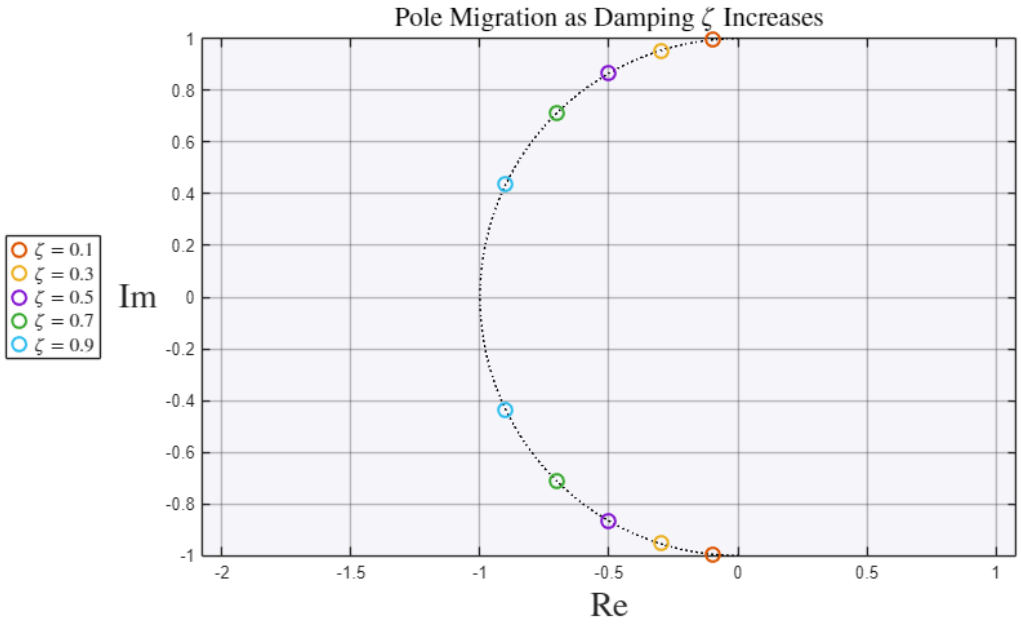
```
info = stepinfo(G)
```

```
info =

  struct with fields:

        RiseTime: 0.4098
    TransientTime: 2.0190
     SettlingTime: 2.0190
      SettlingMin: 0.9315
      SettlingMax: 1.1629
        Overshoot: 16.2929
       Undershoot: 0
             Peak: 1.1629
         PeakTime: 0.8980
```

What Changes with Damping: Pole Locations

The family of step responses at the top is driven by where the poles sit. At fixed wn, increasing zeta slides the complex pair along a circle of radius wn – from the imaginary axis (undamped, sustained ringing) toward the negative real axis (over-damped, no overshoot). This is the "before/after" of adding damping.

```
wn_fixed = 1;
zetas_pm = [0.1 0.3 0.5 0.7 0.9];
figure
th = linspace(pi/2,3*pi/2,200);
plot(wn_fixed*cos(th),wn_fixed*sin(th),'k:','HandleVisibility','off')
hold on
for z = zetas_pm
    p = roots([1 2*z*wn_fixed wn_fixed^2]);
    plot(real(p),imag(p),'o','MarkerSize',8,'LineWidth',1.5, ...
        'DisplayName',sprintf('$\\zeta=%.1f$',z))
end
hold off
grid on; axis equal
legend('Interpreter','latex','FontSize',11,'Location','westoutside')
title('Pole Migration as Damping $\zeta$ Increases','Interpreter','latex','FontSize',16)
ylabel('$\mathrm{Im}$','Interpreter','latex','FontSize',20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$\mathrm{Re}$','Interpreter','latex','FontSize',20)
```



Try it yourself

- Set `zeta = 0.1` and notice the huge overshoot and long ringing; then try `zeta = 1.2` for a sluggish, overshoot-free response.
- Hold `zeta` fixed and double `wn`: notice the shape stays the same but the time axis compresses (faster, same overshoot).